Chaos Theory: Delightful Unpredictability

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1 Introduction

There is a scene in the movie Jurassic Park in which the fictional character Dr. Ian Malcolm attempts to explain the theory of chaos. He tells Dr. Settler that "it simply deals with predictability in complex systems. The shorthand is the butterfly effect, the butterfly flaps its wings in Central Park, you get rain in central Asia." Then, he illustrates further by dropping water droplets on her hand and saying that the flow of the water will change in unpredictable ways every distinct time "because tiny variations... the orientation of the tiny hairs on your hand, the blood cells and imperfections in the skin."

To a layperson, the concept of chaos conjures up images of randomness, changes, and unpredictability. In scientific circles, it means fluctuations in a deterministic system so sensitive to precise measurements that the output may appear completely random, yet with an exquisitely patterning structure. In this article, we will take a walk through this fascinating subfield of dynamic systems.

2 Advances in Chaos Theory and its Mathematics

Within the mathematical scene, progress is often made by intersecting parts of disciplines, revitalizing and building on precursors. Revolutionary mathematical ideas come from lots of hard work and collaboration with others. That's true of the evolution of chaos theory, too. The beginning of the story of chaos theory can be traced to the effort of finding a solution to the motion of the three-bodies, the problem set in a branch of physics following the gravitational relationships between celestial bodies. Newton who extensively studied the motion of the Earth around the sun was able to describe any system in which motion results from applied forces is a dynamical system, in the sense that its behavior evolves in time. Since this fundamental principle of dynamics establishes a relation between an acceleration and the applied forces, it is expressed by a differential equation which involves the second derivative of the position — the acceleration — and the position evolving under the influence of the forces on which it depends. Newton's solution to the problem of the motion of the Earth around the sun (law of gravitational attraction) naturally invited scientific attempts to extend Newtonian formulas to the sun, Earth and moon.

These efforts proved fruitless, since the three-body problem did not have a closed solution and small variations in orbital influences drastically changed movement. Ultimately, the seminal work of the French mathematician Henri Poincaré in the 1800s developed an insightful approach to the problem. The challenge lied in nonlinearity. Unlike linear systems that can be subdivided into smaller parts and combined back together (which allows for the simplification of complex problems), nonlinear systems do not fulfill the requirements of the superposition principle. The richness of behaviors that we encounter in our daily lives cannot be explained by a mathematical relationship of proportionality. For example, the spread of infectious diseases is heavily reliant on time and dependent on its initial variables, and thus in the long term, prediction is impossible. In the real world, it is difficult to find appropriate, predictable linear behavior for such natural processes. Poincaré saw that a system with even only three bodies would behave too unpredictably to be modeled using conventional methods. He opted for applying pictures and geometry to complex problems, thus introducing tools for dealing with dynamics. Thus, the mathematician's observations gave rise to the field of dynamic systems and offered a glimpse into the possibility of chaos.

The top floor of MIT's Green Building houses the Lorenz Center; it is named after Edward N. Lorenz who made a serendipitous discovery that subsequently altered the course of chaos theory and broke into popular culture. By the early 1960s, Lorenz had used a computer to construct a mathematical model of the weather, effectively combining the fields of mathematics and meteorology. His simulations used a set of differential equations which represented long-term changes in temperature, pressure, wind velocity, etc. Lorenz kept a continuous simulation and, as the story goes, one fateful day, he took a coffee break while it ran. Upon his return, he noticed that the weather patterns of the simulation changed drastically when compared to the previous ones. Lorenz initially thought there was an issue with the machine, but upon looking at the data, he realized that the differences stemmed from a simple rounding issue; Lorenz had inputted .506127 as .506. This small variation led to an output which did not uphold the previous results. The system was governed by deterministic laws: with finite systems of differential equations hence, with no randomness invading between one moment and the next. When the scientist narrowed the number of equations from twelve to three, Lorenz observed that the solutions kept oscillating, in an aperiodic state. This phenomenon, called sensitive dependence on initial conditions, rendered the solutions unpredictable. Lorenz illustrated this idea using an analogy of the mere flapping of butterfly's wings that can change the weather.

In his landmark work "Deterministic Nonperiodic Flow," Lorenz also showed that there was an underlying structure in chaos. While graphing data in three dimensions, he observed that plotting the trajectory of two nearby points would create ever-growing and widely separated regions, but iterating points off the regions would bring them closer. Such a complex system is called a "strange attractor," and dynamics discovered by Lorenz called the "Lorenz attractor."

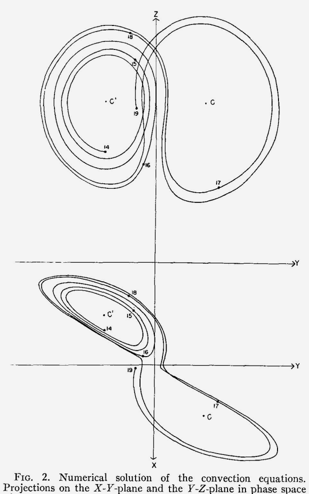


FIG. 2. Numerical solution of the convection equations. Projections on the X-Y-plane and the Y-Z-plane in phase space of the segment of the trajectory extending from iteration 1400 to iteration 1900. Numerals "14," "15," etc., denote positions at iterations 1400, 1500, etc. States of steady convection are denoted by C and C'.

Figure 1. Lorenz' original attractors

It bears mentioning here, that without the advent of computers, the discovery of the attractors would prove to be difficult; and without the work of two female mathematicians, the discovery of Lorenz's attractors would be perhaps impossible. Margaret Hamilton and Ellen Fetter, who programmed, plotted and supplemented Lorenz' research, helped reveal the mathematical signatures of this chaos (later, Hamilton's code became the structural foundation for many of NASA's future programs, including Skylab—the first American space station). Their contributions were indispensable in the field, and Lorenz acknowledged Fetter's help at the end of his original paper.

As a warm-up before taking on the function, let's examine a simple dynamic system that is not chaotic. In the given state s of the system at one moment in time, $f(s)$ is the state after one phase of time has passed. Following the behavior of a dynamic system, inputting the output back into the function will get us $f(f(s))$ which would be fed back in and so on. The dynamics of the system can be written as the following sequence: $s, f(s), f(f(s)), f(f(f(s)))...$ As an example, let's consider the function $f(x) = 2x(1-x); 0 < x < 1$. Let's pick an arbitrary number for x from the given interval, and use 0.75. We get $f(0.75) = 0.375$, then $f(f(0.75)) = f(0.375) = 46875$, and then $f(f(f(0.75))) = f(0.46875) = 0.498047$, followed by $f(f(f(f(0.75)))) = f(0.498047) = 0.499992$. We see that with the initial state at 0.75, the system gets closer and closer to the state 0.5 with every step and passage of time (by the way, we can start with 0.25 for x, or any other value between 0 and 1 and the solution will still gravitate to 0.5).

Now, let's change the function to $g(x) = 4x(1-x)$ with x belonging to the interval 0 to 1. Then, $g(0.75) = 0.75, g(g(0.75)) = 0.75$, and continuing in this manner will reveal that all the states will remain fixed through all iterations. A dynamic state can also loop between two values, or some other finite set of points in any length of loop, or jump wildly without setting. More interestingly, two initial seed values of x within 0.000001 of one another will quickly form different trajectories, and the system will resemble nothing of the previous pattern. This manifests one of the defining features of chaos: sensitivity to initial conditions.

In the early 1960s, the American mathematician Stephen Smale discovered a simple and stable function that provided an explanation to the chaos of Poincare's three-body system. Smale's background in geometric analysis supplemented by his rigorous studies of literature on differential equations led to the hallmark of chaos: Smale's horseshoe function.

To further understand chaotic systems and Smale's horseshoes, let's subdivide our interval depending on whether the initial value is $x < 1/2$ or $x > 1/2$. After the orbit commences, let's note where the next iteration lands on. Subsequently, continue commencing orbits using the previous output as the new input, keeping track of which half of the horseshoe the output falls on. Each sequence of these recorded halves can be traced back to one specific initial input. Our function g, unlike our previous function f, does not express asymptotic behavior, and therefore maps to orbits of every random design. Now we are ready to expand our examples to Smale's horseshoe, used to plot functions with more dimensions. First, we can create a rectangle with vertices A, B, C, and D. We can then horizontally elongate and contract a new rectangle into the shape of a horseshoe.

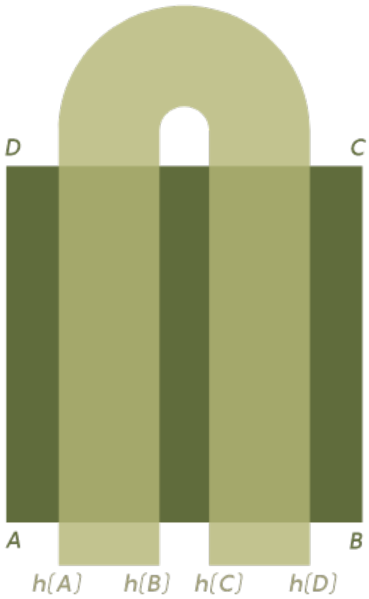


Figure 2. Smale's horseshoe mapped onto $ABCD$

Overlaying this horseshoe figure over rectangle $ABCD$ defines a function h which maps a point h into a point inside the horseshoe. Remapping these points by bending new horseshoes over and over demonstrates the phenomenon of chaos, where the outputs grow increasingly distinct and separate. Notably, the horseshoe is invertible, so mapping the inverse function of h yields a new, perpendicular horseshoe.

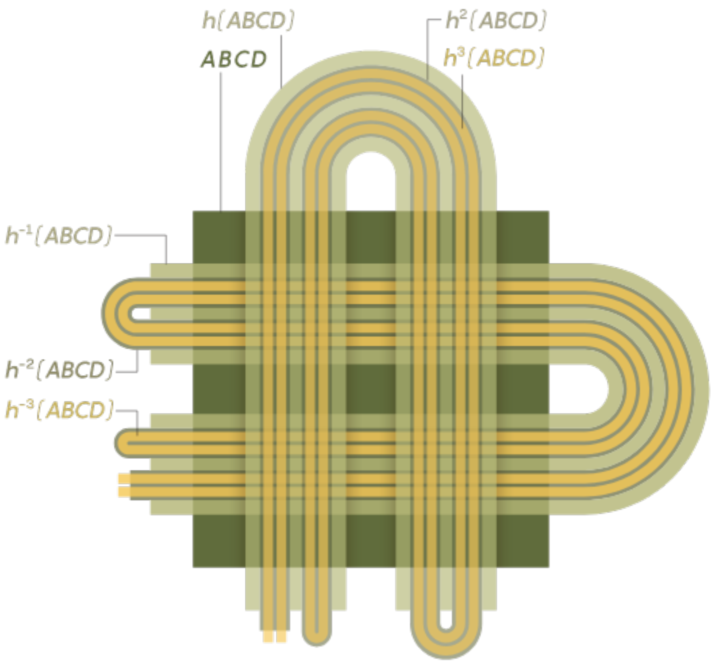
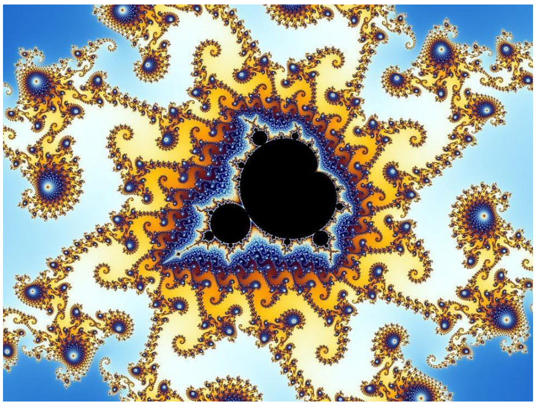


Figure 3. An example of points on a horseshoe being repeatedly mapped

The term *chaos* was introduced in a 1975 paper by James A. Yorke and his student T.Y. Li. The same year French-Polish mathematician Benoit Mandelbrot coined the term fractals as another class of "strange attractors," where smaller regions appear similar in structure to the whole thing.

(Both men received the 2003 Japan Prize in science and technology). Needless to say, fractals with their aesthetic appeal have added to popular fascination with nonlinear dynamics and chaos.



3 Conclusion

In the 1980s, movies such as *Jurassic Park* have helped to stimulate greater public interest in chaos theory. Although the fictional Dr. Malcom's efforts to explain the essence of chaos are rather clumsy, Professor Yorke of the University of Maryland in an interview remembers at least one graduate student in particular who "showed up from Australia because he liked the movie [Jurassic Park] and thought he should study chaos. And he wound up doing that." Perhaps this stretch into popular culture has lessened in the 21st century, but it does not mean that the influence of chaos theory has diminished. Nowadays, applications of the mathematics of chaos are deeply diverse and reach far beyond celestial mechanics. Studies on chaos dynamics extend to heart rhythms, laser physics, and the firing of neurons in our brain. On the philosophical level, chaos theory instructs us that we cannot predict what the future holds, but perhaps walking into chaos can be immensely enjoyable.

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